Reaction wheel sizing: equation summary

Code: `wheel_sizing.py` (equations 1 to 7), `momentum_core.py` (equation 4). Case: spiral-outward assembly, 264 s, ZO-31 design parameters (35.5 kg robot, 0.4375 kg tiles, 30 mm pitch). The first run used the old parameters (309 s; x 1.09, y 1.14, z 7.33 N m s).

Step 1. Momentum balance

B is the structure frame. There is no external torque and the system starts at rest, so total angular momentum stays zero.

$$\mathbf{J}\,\boldsymbol{\omega} + \mathbf{h}_{rel} + \mathbf{h}_w = \mathbf{0} \tag{1}$$

To keep the structure still ($\boldsymbol{\omega} = \mathbf{0}$), the wheels store the opposite of the internal momentum:

$$\mathbf{h}_w = -\,\mathbf{h}_{rel} \tag{2}$$

The motors must supply this torque:

$$\boldsymbol{\tau}_{motor} = \frac{d\mathbf{h}_w}{dt} = -\frac{d\mathbf{h}_{rel}}{dt} \tag{3}$$

The internal momentum is the sum over every moving body i (robot, gantry, tiles):

$$\mathbf{h}_{rel} = \sum_i \left[m_i\,\boldsymbol{\rho}_i \times \mathbf{u}_i + \mathbf{I}_i\,\boldsymbol{\omega}_i^{rel} \right] \tag{4}$$

Step 2. Peak needs

For each body axis $k \in \{x, y, z\}$, take the peak over the whole run:

$$H_{req,k} = \max_t |h_{w,k}| \qquad \tau_{req,k} = \max_t |\tau_{motor,k}| \tag{5}$$

Step 3. Size the wheel

$$H_{design} = n\,H_{req} \qquad \tau_{design} = n\,\tau_{req} \qquad I_w = \frac{H_{design}}{\Omega_{max}} \tag{6}$$

$$m_w = \frac{I_w}{k_s\,r^2} \qquad P = \tau_{design}\,\Omega_{max} \qquad \Omega_{max} = \mathrm{RPM}_{max}\,\frac{2\pi}{60} \tag{7}$$

Symbols

Symbol	Meaning	Value
$\mathbf{J}$	Inertia of the whole system about its centre of mass	Changes with time
$\boldsymbol{\omega}$	Structure angular velocity	Goal: 0
$\mathbf{h}_{rel}$	Momentum of the moving parts, relative to the structure	From the motion plan
$\mathbf{h}_w$	Momentum stored in the wheels	Result
$\boldsymbol{\tau}_{motor}$	Wheel motor torque	Result
$m_i,\ \boldsymbol{\rho}_i,\ \mathbf{u}_i$	Mass, position, velocity of body i relative to the system centre of mass	From the motion plan
$\mathbf{I}_i,\ \boldsymbol{\omega}_i^{rel}$	Own inertia and spin of body i	From the motion plan
n	Safety margin	1.5
Ω_{max}	Maximum wheel speed	6000 rpm
k_s	Wheel shape factor (1 = thin rim)	1.0
r	Wheel radius	0.10 m

Results

Axis	H_{req} (N · m · s)	τ_{req} (N · m)	I_w (kg · m^2)	m_w (kg)	P (W)
x	3.51	12.85	0.0084	0.84	12,112
y	4.01	14.03	0.0096	0.96	13,225
z	6.90	25.55	0.0165	1.65	24,079

Limits

- Rigid, isolated model. No gravity gradient, drag or external torque.
- Wheel mass is not added to **J**.
- Three orthogonal wheels, one per axis.
- P is an upper bound. Peak torque and top speed do not happen together.
- The code does not check that real hardware can deliver this torque and speed.